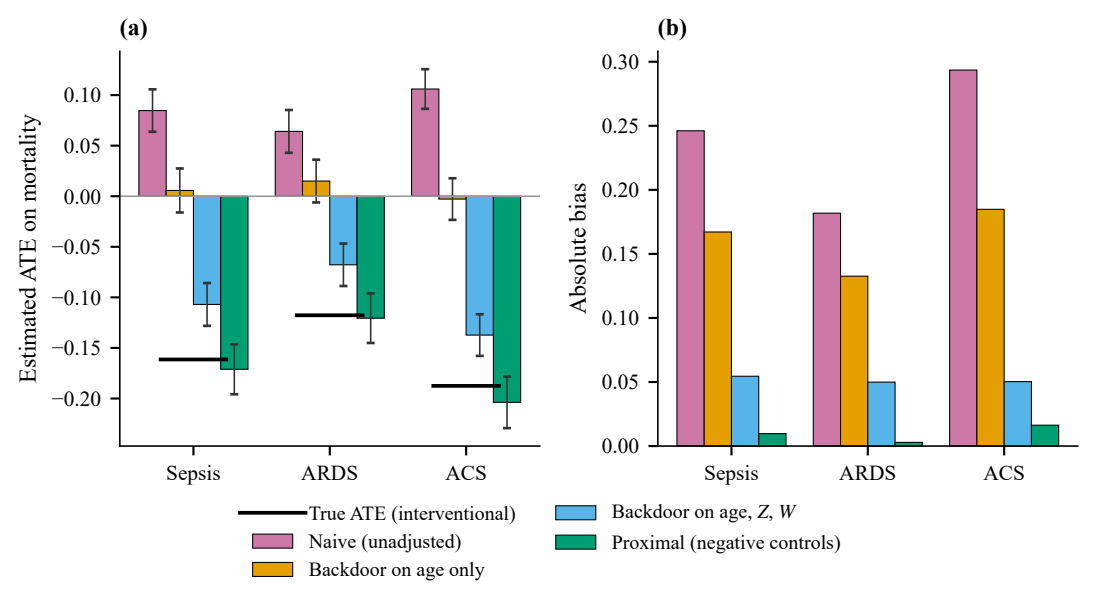

(a)
(b)
Estimated ATE on mortality
0.10
0.05
0.00
−0.05
−0.10
−0.15
−0.20
Sepsis
ARDS
ACS
Absolute bias
0.30
0.25
0.20
0.15
0.10
0.05
0.00
Sepsis
ARDS
ACS
True ATE (interventional)
Naive (unadjusted)
Backdoor on age only
Backdoor on age, Z, W
Proximal (negative controls)